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Biome type and availability of natural vegetation dictate whether farmland intensification or expansion is worse for biodiversity

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Abstract

To reduce the biodiversity impact of agriculture, increasing yields on existing farmland has been proposed as an alternative to farmland expansion. However, the relative effects of yield increases versus agricultural expansion have mostly been examined regionally, and measured in terms of species persistence—a metric relevant to extinction risk but limited in describing ecological communities and their support for ecosystem services. Without a thorough analysis, the lower biodiversity impacts of agricultural intensification remain largely speculative. This study provides a global assessment of biodiversity responses to land conversion and yield increases, including closing yield gaps. We also compare the biodiversity impacts of expanding farmland versus intensifying yields in agricultural landscapes to achieve a 1% increase in total production. Utilizing a large biodiversity database, natural vegetation data, and agricultural yield estimates at the landscape scale, we assess three biodiversity metrics: species richness, total abundance, and relative community abundance-weighted average range-size (RCAR), which provides a proxy for biotic homogenisation. Our models highlight that land conversion is associated with significant biodiversity loss at both local and landscape scales, emphasizing the importance of avoiding farmland expansion into new landscapes. However, yield also lead to significant biodiversity loss; closing yield gaps is associated with a median species loss of nearly 11%, and median abundance loss of almost 13%, with some agricultural landscapes losing almost 90% of species and more than 90% in abundance. Additionally, 30% of global agricultural landscapes, predominantly in the tropics, are likely to experience increased biotic homogenization. Neither expansion nor intensification is consistently better for biodiversity, with biome type, crop, biodiversity metric, and percentage of natural vegetation influencing which approach is less harmful. Our results suggest that minimising the biodiversity cost of agriculture requires a context-dependent balance between intensification and expansion in agricultural landscapes.

Introduction

Agriculture is one of the major drivers of biodiversity loss (Maxwell et al. 2016; Uhler et al. 2021) and many projections suggest that biodiversity loss due to agriculture may continue to occur locally, even under ambitious scenarios that include efforts to limit future demand for agricultural products (Kok et al. 2023; Leclère et al. 2020; Williams et al. 2021). Policies attempting to reduce the biodiversity cost of agriculture often aim to reduce farmland expansion by increasing yields on existing agricultural land (Mueller et al. 2012; Phalan et al. 2014). To understand the effectiveness of this strategy we need to fully quantify the biodiversity impact of yield increases and compare it to the impact of farmland expansion. The intensification-expansion options are often discussed within the land-sparing/sharing framework, in which the two extreme options are land sparing, when production is concentrated on the smallest possible area by intensifying agriculture, and land-sharing, when production is spread with the lowest possible local impact through the use of biodiversity-friendly practices (Balmford 2021). The land-sparing/sharing studies often rely on models of population density in relation to yield at regional scales as a measure of likelihood of species persistence, aimed at understanding extinction risk. Most of these studies conclude that more species would benefit from land sparing than from land sharing when aiming

to reach a given production target (Balmford 2021; Dotta et al. 2016; Kamp et al. 2015; Phalan et al. 2014; Williams et al. 2017) whereas other studies show that species responses are mediated by surrounding natural habitat (Gilroy et al. 2014; Karp et al. 2019; Macchi et al. 2020) and land-use history (Feniuk et al. 2019; Finch et al. 2019; Simons & Weisser 2017). Studies of local community metrics in relation to yield, which is important to understand ecosystem service supply (Winfree et al. 2015), and stability of ecological communities (Hatton et al. 2024) are rarer, and show mixed responses of biodiversity to yield increases (Clough et al. 2011; Guerrero et al. 2012; Winqvist et al. 2014; Wurz et al. 2022). Neither species persistence or community metrics response to yield increases were assessed globally so our understanding of the comparative biodiversity impacts of agricultural intensification and expansion is far from complete.

To quantify the biodiversity impacts of agricultural management, previous large-scale studies have either used broad categories of land-use intensity (which typically consider both agricultural inputs and outputs) (Beckmann et al. 2019; Newbold et al. 2015), modelled proxies such as human-appropriated net primary productivity (Semenchuk et al. 2022), or agricultural inputs such as fertilisers (Semenchuk et al. 2022). As agricultural management is only a means to obtaining a certain level of yield, focusing instead on the biodiversity impact of yield will provide a more direct relationship to consumption patterns, which foregrounds the importance of the demand-side measures to reducing biodiversity impacts of agriculture (Leclère et al. 2020). This approach also provides a more direct relationship to agricultural production as higher agricultural inputs do not always translate into higher yield, and degraded farmland might actually require higher inputs for the same yield level (Ray et al. 2012). Similarly, yields can be increased with lower biodiversity impacts when relying on ecosystem services (Kleijn et al. 2019; Rockström et al. 2017) or using technological improvements benign for biodiversity (Sajjad et al. 2022).

To fully quantify the biodiversity impact of yield increases, we have to look beyond the most frequently-used metrics of species richness (Attwood et al. 2008; Beckmann et al. 2019; Newbold et al. 2015) and total abundance (Attwood et al. 2008; Newbold et al. 2015), which can obscure changes in community composition. Biotic homogenisation, which happens when ecological communities become increasingly similar across space (Baiser et al. 2012), is often a consequence of human impact even when other biodiversity metrics appear unchanged (Dornelas et al. 2014). One metric capturing biotic homogenisation of ecological communities is the community-average range size, often weighted by the abundance of the component species, because more widely-distributed species are part of more ecological communities, making them more similar (Newbold et al. 2018). Apart from Phalan et al. (2011, 2014) who report how global ranges of birds and trees in Ghana and India relate to species' ability to persist in agricultural landscapes at different yield levels, few studies report how biotic homogenisation relates to agricultural management (Newbold et al. 2018) and none to different yield levels at global scale.

Here, we provide a large-scale assessment of the biodiversity consequences of yield increases and land conversion in landscapes producing four crops (maize, soybean, wheat and rice). We also compare the potential biodiversity impacts globally of pursuing these two options to achieve a 1% increase in

production. We achieve this in three stages, each focused on a related question: 1) what is the biodiversity impact of land conversion in the absence of any yield considerations, considering both local and landscape-level effects of land conversion; 2) within agricultural landscapes, what is the biodiversity impact of increasing yields and closing yield gaps in the absence of land conversion; and 3) using the models developed in step 2, how do the potential impacts of land conversion and yield increases compare across global agricultural landscapes for a hypothetical 1% increase in production? For our purposes, we define landscape scale as the extent of one raster cells in our spatial data. We analyse four crops separately to capture potential differences in landscape types and landscape effects of different management approaches. We use a space-for-time substitution to model three biodiversity metrics: sampled species richness, sampled total abundance, and relative community abundance-weighted average range-size (RCAR). Firstly, we model how biodiversity reacts to land-use conversion locally (farmland versus primary vegetation), and at landscape scale (modified landscapes with less than 30% natural habitat versus natural landscapes with more than 70% natural habitat). Secondly, we model biodiversity impacts of yield increases in cropland, and in primary vegetation in the farmed landscape, accounting also for amount of natural vegetation in the landscape (Hoskins et al. 2016), among other factors, in preparation for answering the third question of this study. To illustrate the potential upper bounds of biodiversity effects of yield increases, we use the selected models to calculate biodiversity change weighted by cropland and primary vegetation proportions in the landscape when closing yield gaps as identified based on agricultural estimates for the year 2000. Thirdly, we compare the impact of conversion against yield increases by using the models produced in step 2 to project biodiversity change in two scenarios: one in which farmland cultivated with the respective crop expands by 1% at the cost of primary vegetation in cultivated landscapes, and one in which the agricultural yield increases by 1% on already existing farmland, leading to identical total production increases.

Methods

Biodiversity data

For biodiversity data, we used the 2016 release of the PREDICTS database, which contains 3,250,404 biodiversity records, mostly sampled from 2000 to 2012, from 666 published studies (Hudson et al. 2017). Each study within the PREDICTS database contains data sampled with the same method across a gradient of land use or land-use intensity. The data in each study are grouped into one or more spatial blocks, each containing data from one or more sites. Each site is attributed one of 6 predominant land-use classes based on the information provided in the original papers, or by the authors of those papers. For our purposes, we selected only those sites located in cropland (i.e., land used for cultivation of herbaceous crops) or primary vegetation (i.e., land showing no evidence of prior destruction of vegetation, either by human actions or extreme natural events), resulting in 1,318,867 records from 10,094 sites from 489 studies. The dataset included 18,853 species of which there were 3,994 vertebrates, 6,693 invertebrates, 7,269 plants, 894 fungi, and three protists. For each of the statistical analyses on relative biodiversity of human-modified and natural landscapes, henceforth land-conversion

models, and biodiversity impact of yield, henceforth yield-biodiversity models, we used subsets of these data (Tables S4-S8, Figures S1 – S4), which we selected based on the methods described in 'Data processing for statistical analysis', below.

We used three biodiversity metrics: sampled species richness, total sampled relative abundance, and relative abundance-weighted community-average range size (RCAR). Sampled species richness represented the count of all species sampled at a site, as identified by the authors of the original study (not necessarily always resolving taxonomic synonymy). Total sampled relative abundance was calculated as the sum of abundance measures for all taxa at a site. Abundance was reported in a variety of measures, some of them sensitive to sampling effort. For measures sensitive to sampling effort, and where that effort differed among sampled sites within a study (~ 1% of records), we corrected the raw abundance measure by dividing it by its relative effort as a proportion of the effort of the most sampled site within a study (Newbold et al. 2015). RCAR is a measure of how widely or narrowly distributed are the species in a community, on average. An increase in RCAR metrics indicates a shift towards more widely distributed species on average, which could be caused either by increases in the number or relative abundance of wide-ranging species and/or by decreases in narrow-ranging species. Similarly, a decrease in RCAR indicates a change toward more narrow-ranged species, on average, which could be caused by increases in narrow-ranged species and/or decreases in wide-ranging species. We used a RCAR metric as calculated in Newbold et al (2018) based on occurrence data from the GBIF database (GBIF.org 2015). The estimated area of occupancy of each species, computed as the sum of the total area of grid cells containing GBIF records of the respective species, was extracted at 55 km × 55 km resolution. Modelled responses of RCAR to land use and land-use intensity have previously been shown to be robust to using different grid resolutions to calculate area or occupancy from GBIF records, and also to using alternative measures of range size, including extent rather than area of occupancy (Newbold et al. 2018). Further details of the methodology for calculating RCAR are detailed in Newbold et al (2018).

Model Explanatory Variables

Global agricultural estimates

For the yield-biodiversity models, we used global agricultural estimates based on national and sub-national statistics whose spatial allocation to pixel level was modelled following certain assumptions. Different authors will often use the same reported statistics for the same time periods, but present different spatial patterns due to different modelling choices. To account for the influence that different modelling approaches can have on our results, we use here two global agricultural datasets, EarthStat (Monfreda et al. 2008) and MapSPAM (You et al. 2014), but focus on EarthStat for model selection and visualisations and use MapSPAM for robustness testing. Although MapSPAM has the advantage that it covers three years (2000, 2005 and 2010) instead of only one year (2000 for EarthStat), EarthStat provides information on the resolution (i.e., administrative level of reporting) of the reported agricultural data that influences the quality of the spatial modelling (Monfreda et al. 2008), and which we used to

discard spatial estimates based on statistics of lower quality. MapSPAM also has the disadvantage that it includes biophysical potential of crops as one of the assumptions used for the spatial disaggregation of agricultural data, which risked creating circularity in our statistical models, because we control for agricultural suitability (see Subsistence yield as proxy for agricultural suitability subsection).

EarthStat

EarthStat agricultural estimates include global estimates of yield and harvested areas for 175 crops in 206 countries around the year 2000 at a spatial resolution of 5’ × 5’ (~ 10 km × 10 km at equator) (Monfreda et al. 2008). The EarthStat authors preferentially collected subnational agricultural statistics at one (state or province) or two administrative levels (county or district) below the national level, where available. Consequently, data were collected from 2,299 administrative units one level below national from 150 countries (72.8% of the total) and 19,751 units two levels below national from 73 countries (35.4% of the total), with data sometimes available at different subnational levels for different areas of the same country. The largest data source of subnational agricultural statistics was Agro-MAPS (George 2004). Other data sources were national census agencies and agricultural surveys for the largest countries such as United States, Brazil and China (Monfreda et al. 2008). In the absence of subnational statistics, Monfreda et al. (2008) used national-level data from the Food and Agricultural Organisation (FAO, 2006), and independent national level data from Afghanistan, Iraq, Somalia, and Taiwan that were not present in the FAO database. A single value for the lowest available administrative unit was calculated by averaging all the information available for the years between 1997 and 2003. The average reported statistics were then disaggregated spatially onto a gridded map according to the fraction of each cell covered by cropland. Further information regarding the methodology for generating the EarthStat estimates can be found in Monfreda et al. (2008) and Ramankutty et al. (2008).

EarthStat data include spatial layers on yield in tons per hectare, data quality for yield information, total production in tons, fractional harvested area, harvested area in hectares, and data quality for harvested area for each crop in each grid cell. For our analysis, we used yield, data quality for yield information, and the harvested area in hectares layers. Given the widely variable data quality and availability across crops and countries, we selected four of the major crops for which estimates are likely to be more reliable (Monfreda et al. 2008): maize, soybean, wheat and rice, which together represent over 60% of total global calorie production (Tilman et al. 2011). The crops were selected also based on availability of subsistence yield data to be used as a proxy for agricultural suitability. We selected for our analysis the EarthStat estimates that relied on subnational level data at one and two administrative levels below national level (top two data quality scores in the yield data quality spatial layer), disregarding estimates based on national-level and FAO statistics.

MapSpam

MapSPAM agricultural estimates were created with the Spatial Production Allocation Model (SPAM), a model produced by the International Food Policy Research Institute. SPAM uses a range of variables to guide spatial allocation (e.g., probability of crop existence in a spatial unit, biophysical potential, area

constraints, management constraints) to guide the spatial allocation of agricultural data into a global 5’ x 5’ (~ 10 km x 10 km at equator) gridded representation of cultivated area, production quantity and yield under different management regimes for the years 2000, 2005 and 2010 (You et al. 2014). Similarly to EarthStat data, MapSPAM relies on sub-national reported statistics sourced from Agro-MAPS and on additional data collected by authors through research partnerships. The sub-national crop data of MapSPAM were collected from 2,758 administrative units one level below national (compared to 2,299 for EarthStat) and 21,498 units two levels below national (compared to 19,751 for EarthStat). Where in disagreement, the sub-national data were calibrated to agree with the national FAO data (You & Wood 2006). For the rest of the area, FAO national statistics were used. MapSPAM estimates do not include spatial information on resolution of the underlying agricultural data and therefore, it was not possible to select spatial estimates based on the highest resolutions. For the year 2000, estimates were available for 20 crops, while all other crops were included in an additional catch-all group. For the years 2005 and 2010, the estimates include 42 crop categories including the catch-all group. The reported statistics used for each time step were averaged over the three years around the reference year (for example, 1999–2001 for year 2000) (Wood-Sichra et al. 2016). Through a combination of data and expert opinion, MapSPAM authors further disaggregate agricultural estimates into four levels according to farming technology and crop management systems: irrigated, rainfed high inputs, rainfed low inputs and rainfed subsistence. We used MapSPAM subsistence estimates, which assumes rainfed conditions and no inputs, to check for the robustness of models to choice of subsistence yield.

The SPAM modelling approach combines several types of data (e.g., revenue, market access models, evidence of existing crop distribution, biophysical suitability, information on agricultural management) to create an informed spatial prior covering the territory over which the agricultural statistics are disaggregated (You et al. 2014). While the EarthStat approach uses only land cover information to allocate the reported statistics, the MapSPAM model uses this informed spatial prior together with land cover to allocate agricultural area and production.

Percentage of natural habitat

We included percentage of natural habitat in our models because of previous studies indicating the importance of this landscape characteristic on its own (Outhwaite et al. 2022) and in interaction with yield (Macchi et al. 2020) for determining biodiversity patterns. We extracted for each data point the percentage of natural habitat at that location by using a modified version of the land-use estimates from Hoskins et al. (2016). The original dataset at 30-arc-second resolution includes estimates of proportional cover for five land-use classes (urban, cropland, pasture, secondary habitat and primary habitat). These estimates were obtained by downscaling the Land-use Harmonization dataset (Hurtt et al. 2011), a global time series of land use at a resolution of 0.5° that covers the time period 1500–2100. Hoskins et al. (2016) downscale estimates for 2005, based on statistical relationships with fine-scale data on climate, land cover, landform, human population density and accessibility defined at the level of 61 unique combinations of biomes and biogeographical realms. The year 2005 is within the sampling timespan of the PREDICTS data and therefore, a suitable data source. We projected primary and

secondary vegetation proportional cover estimates onto a Behrmann equal-area projection, with 1-km spatial resolution, using the projectRaster function with bilinear interpolation in the raster R package, and then summed them to obtain an estimate for natural habitat cover. We then calculated the average percentage cover of natural habitat within 5-km × 5-km grid cells using the 'zonal' function in the raster package.

For the yield-biodiversity models, we used percentage of natural habitat as a continuous variable and interaction term with yield, land use, biome type and subsistence yield, to account for potentially different biodiversity responses to these variables at different levels of natural habitat (Macchi et al. 2020; Outhwaite et al. 2022). For the land-conversion models, we used percentage of natural habitat to select sites that were located in pixels with less than 30% natural vegetation ('human-modified landscapes'), and pixels with more than 70% natural vegetation ('natural landscapes') (Table S4). We chose this threshold of natural habitat based on previous literature showing that biodiversity loss becomes steeper as landscapes lose more than 30% of suitable habitat (Andren 1994). We then combined this landscape classification based on natural vegetation with the PREDICTS land-use classification to create a combined variable describing landscape modification and local land use in four groups: primary vegetation in natural landscapes, primary vegetation in human-modified landscapes, cropland in natural landscapes, cropland in human-modified landscapes.

Land use

Primary vegetation is the land-cover type that preserves best the biodiversity in a landscape and often represents a source of biodiversity-based ecosystem services in agricultural landscapes (Karp et al. 2013), but it is also impacted by neighbouring agricultural expansion and management (Rand et al. 2006). To capture potential differences in the impact of yield increases on cropland and primary vegetation within the same landscape, we included the PREDICTS land-use classification as a variable in the models. For the yield-biodiversity models, we used the original land-use classification from the PREDICTS database and interaction terms between land use, and yield and percentage of natural vegetation as we expect that these effects will differ depending on local land use (Outhwaite et al. 2022). For the land-conversion models, we used the land-use classification in combination with percentage of natural habitat, as detailed above.

Biome type

We included a classification of biome type in our models because the different biodiversity patterns and socioeconomic conditions across world biomes is likely to impact the biodiversity response to land use and agricultural management (Newbold et al. 2020). To improve the tractability of statistical models compared to using the usual 14 biomes classification (The Nature Conservancy 2009), we assigned each data point a tropical/non-tropical classification based on their latitude. Data points located between 23.5°N and 23.5°S were identified as tropical, and otherwise as non-tropical. For the yield-biodiversity models, we also included interaction terms between biome type and both yield and percentage of natural vegetation, because these two variables could have different biodiversity impacts

depending on the environmental and socioeconomic differences captured by the tropical/non-tropical classification.

Subsistence yield as proxy for agricultural suitability

The suitability of different areas for growing crops varies widely, resulting in wide variation in the inputs required to achieve similar yield levels, which might affect biodiversity for a given level of achieved yield. For example, yields reached with irrigation in water limited regions could be comparable to rainfed yields reached in similar but not water limited regions: however, the management footprints – and therefore the biodiversity impacts – could differ. Moreover, agricultural suitability can correlate positively with different biodiversity metrics because often conditions that are favourable for crops are also favourable for many other species (Chown et al. 2003; Luck et al. 2010), distorting the relationship between biodiversity and yield. We used yields in the absence of inputs, irrigation and most human management, i.e., subsistence yields, as a proxy for agricultural suitability for each crop. We used subsistence yields modelled by a version of the Environmental Policy Integrated Climate (EPIC) model, which was initially developed in the United States to model how soil productivity is affected by agricultural management (Williams et al. 1989). Since then, it has been expanded to allow the simulation of yields and agricultural processes at global and regional scales, and their interactions with agricultural management (Balkovič et al. 2013; Izaurralde et al. 2006; Müller et al. 2019). EPIC uses a daily time step and includes components on crop growth, weather simulation, competition, water, carbon and nutrient cycling, and soil processes. Simulated potential plant growth depends on conversion of CO_2 to biomass, intercepted solar radiation and vapour pressure deficit, while stresses from temperature, nutrient deficit, salinity and other soil factors lead to decreased plant growth. We used here a version of the EPIC model developed at University of Natural Resources and Life Sciences (BOKU) in Vienna, Austria, referred to in the literature as EPIC-BOKU (Müller et al. 2018), which was specifically created to estimate yields at different management intensities (Frank et al. 2015; Havlík et al. 2011; Skalskỳ et al. 2008). We used yield simulated with EPIC-BOKU under a subsistence management system without fertilisation or irrigation at 0.5° x 0.5° resolution. The simulation used the WFDEI meteorological forcing data, which represents historical daily-resolution climate data on land at 0.5° × 0.5° resolution based on ERA-Interim reanalysis data (Weedon et al 2014). For analysis with the EarthStat yield estimates for the year 2000, we averaged EPIC-BOKU subsistence yield estimates between the years 1997–2003. Considering that agricultural suitability could have different relationships to biodiversity at different landscape compositions and yield levels, we added in our models interaction terms between subsistence yield and both yield and percentage of natural habitat. For models with MapSPAM, we averaged EPIC-BOKU subsistence yield estimates between the years 1997–2003 for the 2000 yields, 2002–2008 for the 2005 yields, and 2007–2013 for the 2010 yields. We used a five-year averaging window for MapSPAM as we did for EarthStat in order to have subsistence yield values consistent with those in the Earthstat models when performing the robustness tests described below.

We tested for the robustness of our statistical models to choice of subsistence yield by replacing EPIC-BOKU subsistence yield with MapSPAM estimates of subsistence yield in both the Earthstat-based and

MapSPAM-based yield-biodiversity models. MapSPAM estimates of subsistence yield were obtained through a combination of data and expert opinion that allowed authors to disaggregate reported agricultural statistics into four levels according to farming technology and crop management systems: irrigated, rainfed high inputs, rainfed low inputs and rainfed subsistence. For a few large countries management information was collected at the level of the first subnational administrative level below national (You et al. 2014). The authors collected information at national level on the ratios of yields under irrigated and rainfed conditions, and under high-input rainfed and low-input rainfed conditions for each crop. They then use these two ratios and areas under different production systems to calculate estimates of yield under different management systems, using reported yields as average yields across all the production systems within a pixel.

Other variables

To control for additional factors that could influence biodiversity outcomes in the yield-biodiversity models, we extracted a number of spatial variables at the locations of our biodiversity data to be included in the statistical models. Crop diversity has been found to sometimes have a positive effect on local biodiversity (Palmu et al. 2014). We therefore calculated a Shannon crop diversity index based on all EarthStat estimates for all 175 crops:

$$H' = -\sum_{i=1}^{n} p_i \ln p_i$$

3

, where H' is the Shannon index, p_i is the proportion of cropland covered by crop i out of the total cropland for all n crops in the pixel. The EarthStat estimates were more appropriate for this calculation as Earthstat data covers a greater number of crops than MapSPAM.

To account for potential effects of agricultural management and expansion at a larger scale than the pixel scale, we used percentage of natural habitat, as well as yield data from EarthStat and MapSPAM, to calculate an average percentage of natural habitat and yield in a one-pixel buffer (~ 10-km radius) around the focal pixel where the PREDICTS site was located. Because the average yield/percentage of natural habitat in the buffer was highly correlated with the value in the focal pixel, we calculated and used in the statistical models the difference between the buffer and the focal pixel value (buffer-focal difference). The yield buffer-focal difference calculated with EarthStat was used in the yield-biodiversity models with EarthStat data and the yield buffer-focal difference calculated with MapSPAM was used in the robustness checks with MapSPAM data.

To account for the duration of substantial human landscape modification, we used the HYDE model (Klein Goldewijk et al. 2011) to calculate the number of years since 30% of a grid cell (0.5° × 0.5° resolution) is estimated to have become covered by human land uses. We then extracted this information at the location of the biodiversity data. Finally, we also extracted climate information at the

location of the biodiversity data, based on the WorldClim estimates of historical annual mean temperature and annual precipitation for the period 1970–2000 (Fick & Hijmans 2017). We added climate variables in our models because climate is an important determinant of biodiversity levels. However, climate might also correlate strongly with some of our models variables (e.g, yield, subsistence yield), in which case the respective climate variable was removed.

We log_e-transformed variables with right-skewed distributions: annual precipitation, MapSPAM yield, and MapSPAM subsistence for all crops, EPIC-BOKU subsistence for maize and wheat, and EarthStat yield for rice. We added a value of 0.01 (less than 1% of the highest value) to deal with 0 values for MapSPAM subsistence yields for all crops, and for EPIC-BOKU subsistence-yield estimates for maize. Variables with left-skewed distributions were transformed with the formula:

$$V_t = log_e(C - V)$$

,

where V_t is the transformed variable, C is a constant used to avoid negative values, and determined by adding a small number to the highest value of the respective variable and rounding it up. Variables with a left tail distribution were duration of substantial human landscape modification for all crops (with C = 2005, obtained by rounding up the sum between maximum value 1981.6 and 20) and annual mean temperature for rice (with C = 30, obtained by rounding up the sum between maximum value 28.5° and 1°). We then scaled all continuous variables to have a mean of 0 and a standard deviation of 1.

Data processing for statistical analysis

All spatial estimates were transformed to the Behrman equal-area projection and resampled to the EarthStat resolution and extent in *R* (R Core Team 2022) with the *projectRaster* and *resample* functions, respectively, from the *raster* package, version 3.6–23 (Hijmans 2023). We used bilinear interpolation for spatial averaging. We then extracted the values within spatial layers at the locations of the PREDICTS sites based on geographical coordinates. For the yield-biodiversity models for each crop, we selected all PREDICTS sites in primary vegetation or cropland that fell within pixels that contained cultivated areas of the respective crop according to EarthStat estimates (at the highest or second-highest data quality level; see Global Agricultural Estimates subsection). The PREDICTS cropland classification does not differentiate according to the crop cultivated at the precise sampling location. Therefore, the biodiversity data were sampled in landscapes used to grow the focal crops (according to EarthStat estimates), even if the actual sample may have been taken in a different crop. In doing so, we assume that intensification levels tend to be similar across cropping systems within an agricultural landscape.

For the MapSPAM yield, we chose either the 2000, 2005 or 2010 estimates based on which of the three years was closest to the year of the midpoint of the collection dates for each PREDICTS site. If a midpoint year was equally close to two MapSPAM years, we chose the earlier year because that would be more likely to represent the agricultural management impacting biodiversity by virtue of preceding in time.

For the land-conversion models, we used sites that were either in natural landscapes (> 70% natural vegetation) or human-modified landscapes (< 30% natural vegetation), a total of 5,328 sites (Table S1). For yield-biodiversity models, the spatial intersection between PREDICTS data and EarthStat estimates resulted in 4,862 sites for maize; 2,404 sites for soybean; 3,227 sites for wheat; and 2,810 sites for rice (Tables S5 - S8) with at least species richness data.

Statistical analysis

We tested for correlations among all explanatory variables for both the yield-biodiversity and land-conversion models. Collinearity among variables was generally low with most correlations between variables being very weak (0 to 0.2) or weak (0.2 to 0.4). A few pairs of variables were moderately correlated (0.4 to 0.6) and a maximum of one pair of variables were strongly correlated (0.6 to 0.8) for some of our statistical models (Table S6). When a pair of variables was strongly correlated, we eliminated one of them (Table S6), prioritising yield, percentage of natural habitat and subsistence yield measures (in this order) to be kept for modelling. In all cases, it was climate variables (annual mean temperature and annual precipitation) that had to be removed due to high correlations with EarthStat yield, Epic-BOKU subsistence yield or, in the case of the yield-biodiversity model for rice, with percentage natural habitat.

We modelled separately the three biodiversity metrics – species richness, abundance and RCAR – using mixed-effects models with random effects for study identity and spatial block nested within study, to account for non-independence of sites. The random effect for study accounted for differences in sampling methods, sampling effort, focal taxonomic group and broad geographic regions among the different studies in PREDICTS. The random effect for spatial block accounted for the spatial structuring of sampling within a study. To account for overdispersion, we included a random effect for site in the models for species richness (Rigby et al. 2008). To reduce the right skew in the abundance data due to measurements across many taxonomic groups and with a variety of methods, we scaled total abundance by dividing values by the maximum abundance within each study (Outhwaite et al. 2022a). We then loge-transformed the rescaled values to further reduce the skew of the distribution adding 0.01 to deal with 0 values.

For the yield-biodiversity models, each data point/site was weighted in the statistical analysis by the area of the respective crop as a fraction of all crops grown in the pixel to account for the differences in crop composition of agricultural landscapes according to the formula:

$$RA_C = \frac{A_C}{\sum_{i=1}^{175} A_i}$$

1

where RA_C is relative area for crop C, where C is maize, soybean, wheat or rice, A_C is the harvested area in hectares for crop C, and A_i is the harvested area for each of the 175 crops available in the EarthStat

dataset. We rescaled the area fractions such that the average weight is equal to 1, to avoid an apparent reduction in total sample size, which would have an adverse influence on some statistical properties:

$$Rescaled_RA_C = N_{Sites} \frac{RA_C}{\sum_{j=1}^{N_{Sites}} RA_j}$$

2

where N_{Sites} is the total number of sites resulting from combining the biodiversity and agricultural data for each crop. We rescaled the area fractions again for each crop and biodiversity metric because the rescaled weights are dependent on the total sum of relative areas, which changes with number of data points.

The default set of initial variables for the land-conversion models was: annual mean temperature, annual precipitation, biome type, and the combination of landscape conversion and local land use. The default set of initial variables for the yield-biodiversity models was: EarthStat yield, yield buffer-focal difference, percentage of natural habitat, percentage of natural habitat buffer-focal difference, EPIC-BOKU subsistence yield, duration of substantial human landscape modification, crop diversity, annual mean temperature, annual precipitation, biome type (tropical/non-tropical), and land use as identified in the PREDICTS database. This set of variables was modified by removing correlated variables for both type of models, as described above (Table S6). For all yield-biodiversity models, we initially included six 2-way interactions: between EarthStat yield and each of the variables EPIC-BOKU subsistence yield, percentage of natural habitat, land use and biome type; and between percentage natural habitat and each of the variables EPIC-BOKU subsistence yield measure, land use and biome type (justifications for considering these interactions were described above). For the soy RCAR model, due to model convergence issues we had to remove 3 of the 6 interactions: EPIC-BOKU subsistence yield interactions with yield and percentage of natural vegetation, and the interactions between biome type and percentage of natural vegetation, which we considered having less support in the literature. Annual mean temperature was also removed as a main effect from the initial model structure, in addition to annual precipitation, which was removed due to a high correlation with EPIC-BOKU subsistence yield. The convergence issues were probably triggered by a reduction in data points (1,482 RCAR data points compared to 2,404 for species richness).

For both yield-biodiversity and land conversion models, we modelled species richness using generalised linear mixed-effects models with a Poisson distribution, while for ($\log_e$-transformed) abundance and ($\log_{10}$-transformed) RCAR we used linear mixed-effects models. We performed backward stepwise selection based on likelihood-ratio tests with a p-value threshold of 0.05 to select the fixed-effects structure for each model. The selected model structures are presented in Table S7. As an alternative way to deal with overdispersion, we reran the species richness models substituting the Poisson distribution with a negative binomial distribution and dropping the random effect for site (Table S20). To test the consistency of our results, we also ran the final model structures using a Bayesian modelling framework

(Table S21). These models using uninformative priors used 4 Markov chains and 5000 iterations with 2500 warmup iterations. We considered that the models converged if Rhat convergence diagnostic was ≤ 1.01 and rank-normalized effective sample size was ≥ 400, which represents 100 times the number of chains (Stan Development Team 2024).

We conducted all statistical analyses and projections of biodiversity impacts of yield increases in R 3.6.3 (R Core Team 2022). For the mixed-effects modelling, we used the glmmTMB package, version 1.1.5 (Brooks et al. 2017). For model selection, we adapted several functions from the packages associated with the PREDICTS data (Newbold 2018). For checks of the models assuming a Poisson distribution we used *DHARMa* package (Hartig 2022), and for the Bayesian models we used the *brms* package (Bürkner 2021).

Projections of the impact of closing yield gaps

We used the best EarthStat–EPIC-BOKU models to project biodiversity change when closing yield gaps for the global areas occupied by the four crops in the year 2000. Yield gaps where defined by Mueller et al. (2012) as the difference between attainable yields, which are the area-weighted 95th percentiles of observed yields within zones of similar annual precipitation and growing-degree days, and yields given by the EarthStat estimates for the year 2000 that are below attainable yield values for the area. We first projected biodiversity levels in cropland and primary vegetation for yield levels in the year 2000. We then used the same models to project biodiversity levels in cropland and primary vegetation at yields equal to those necessary to close yield gaps. We then calculated an aggregated estimated biodiversity value across all four crops for each pixel, for each metric at each of the two yield levels, using the following formula:

$$BD = \sum_{i}^{n=4} p_i (BD_{Cropland} p_{Cropland} + BD_{PV} p_{PV})$$

3

where *BD* is the biodiversity metric in the respective pixel, p_i is the proportion of crop *i* in the total cropland of the respective pixel, $BD_{Cropland}$ and BD_{PV} are values of the biodiversity metric estimated by the yield-biodiversity model for cropland and primary vegetation, respectively, and $p_{Cropland}$ and p_{PV} are proportion of each land use type out of the total area of the pixel. Where land use was not selected in the fixed structure of the best models, *BD* was identical for both primary vegetation and cropland. We calculated the percentage change in biodiversity relative to the biodiversity corresponding to the year 2000 yields by subtracting in each pixel the biodiversity metric at gap-closing yields from the biodiversity metric corresponding to year 2000 yield and dividing the difference by the biodiversity metric corresponding to year 2000 yields. The biodiversity contribution of the other land uses in the pixel are assumed to be constant at the two time steps and therefore, it does not influence percentage change in biodiversity.

Projections of the impact of cropland intensification vs. expansion into primary vegetation

Cropland expansion can take place at the cost of most land-uses present in a landscape at a given time. Cropland expansion into other land-uses than primary vegetation (e.g., pastureland) is likely to lead to lower biodiversity impacts. We consider here cropland expansion at the cost of primary vegetation to establish a “worst-case scenario” against which to compare intensification. It is important to note that our scenarios for achieving a 1% increase in production exclusively through either intensification or cropland expansion into primary vegetation are not realistic projections for the future but rather extreme cases of a range of possible strategies. However, this scenario exercise allows us to contrast the biodiversity outcomes of intensification and expansion as two extreme approaches to increasing agricultural production.

To compare cropland expansion into primary vegetation to intensification within cropland, we used the EarthStat–EPIC-BOKU models to project biodiversity metrics for two scenarios that would yield identical increases of total production in each pixel for each crop. We then aggregated the biodiversity metric in each pixel for each crop according to the formula:

$$BD = (BD_{Cropland} p_{Cropland} + BD_{PV} p_{PV})$$

4

where BD is the biodiversity metric in the respective pixel, $BD_{Cropland}$ and BD_{PV} are the model-estimated values of the biodiversity metric for cropland and primary vegetation, respectively, and $p_{Cropland}$ and p_{PV} are proportion of each land-use type out of the total area of the pixel.

For the expansion scenario, we kept the yield equal to the year 2000 yield but increased the area cultivated with the respective crop by 1% at the cost of primary vegetation in the pixel, thus changing $p_{Cropland}$ and p_{PV} in favour of a larger $p_{Cropland}$. This meant also a decrease by the equivalent area in the total percentage of natural habitat variable in the respective yield-biodiversity model. For the intensification scenario, the only variable that changed was the yield for each crop, which increased by 1%. We then calculated a percentage change in biodiversity by extracting the biodiversity calculated for the intensification scenario from the biodiversity calculated for the expansion scenario and dividing the difference by the biodiversity metric corresponding to year 2000 yields and landscape composition. The biodiversity contribution of the other land uses in the pixel are assumed to be constant between the two scenarios and equal to the biodiversity of these land uses corresponding to the year 2000 yield and landscape composition.

Results

Biodiversity impacts of land conversion

Relative to primary vegetation in natural landscapes, all other habitat-landscape categories showed reductions in species richness (Fig. 2a.) and abundance (Fig. 2b.), and increases in relative abundance-weighted community-average range size (RCAR) (Fig. 2c.). Species richness is estimated to decrease by -19.3% [95% CI: -26.8%, -11%] in primary vegetation in modified landscapes, by -26.6% [95% CI: -32.2%, -20.5%] in cropland in natural landscapes, and by -31.8% [95% CI: -39.1%, -23.7%] in cropland in modified landscapes, relative to primary vegetation in natural landscapes. Abundance is estimated to decrease by -37.5% [95% CI: -48.2%, -24.6%] in primary vegetation in modified landscapes, by -30.5% [95% CI: -41%, -18.3%] in cropland in natural landscapes, and by -56.7% [95% CI: -64.9%, -46.7%] in cropland in modified landscapes. RCAR is estimated to increase by 33.9% [95% CI: 9%, 64.6%] in primary vegetation in modified landscapes by 90.1% [95% CI: 65.4%, 118.5%], in cropland in natural landscapes, and by 102.4% [95% CI: 59.3%, 157.2%] in cropland in modified landscapes.

Biodiversity impacts of yield increases

Biodiversity change with yield increases is mediated by land use, biome type and percentage of natural habitat in the landscape. Yield increases were associated most often with decreases in species richness (Fig. 3a, c, d), except in landscapes used to grow soybeans, where soybean yield increases were not associated with any changes in species richness(Fig. 3b), and in landscapes used to grow rice in which there is a high percentage of natural vegetation, where rice yield increases were associated with increases in species richness (Fig. 3d). The association between yield increases and total abundance is more complex (Fig. 3e, f, g, h), the interaction with biome type becoming important to explain abundance patterns within landscapes with maize, soybean and rice cultivation (Figure e, f, h, respectively). The general pattern is that total abundance increases with yield in tropical landscapes and decreases in temperate landscapes, while high levels of natural vegetation accelerate increases and dampen decreases in tropical and non-tropical landscapes, respectively. In landscapes used to grow wheat, total abundance decreases with increases in yield, most strongly in landscapes with a high percentage of natural vegetation (Fig. 3g). Biome type impacts the effects of maize and soybean yield on RCAR, with tropical landscapes experiencing strong negative biodiversity impacts of increasing yields (i.e., steep increases in RCAR), and non-tropical landscapes showing smaller impacts or even decreasing RCAR with increasing yields (Fig. 3i, j). In landscapes with rice cultivation, RCAR decreases with increasing yield where the percentage of natural vegetation is high, but increases in landscapes with a low percentage of natural vegetation (Fig. 3l). Wheat yield was not associated with any change in RCAR. The models were robust to the choice of subsistence yield and fairly robust to the choice of yield data (Tables S10-S19). Most models using yields from MapSPAM agreed with the models that used yields from EarthStat in direction, or in both direction and magnitude of biodiversity change (Table S9). The only MapSPAM yield models that did not agree on the direction of the effect of yield on biodiversity were rice-total abundance and maize-RCAR models (Tables S16-S17).

Among the other variables, subsistence yield was selected in 7 of the 12 models (4 crops X 3 biodiversity metrics), had both positive and negative effects on biodiversity, and interacted with yield and percentage of natural habitat, especially in the total abundance models. All the other variables were

selected in 4 or less of the 12 models. Crop diversity had a negative effect on species richness and abundance in maize landscapes but it had a positive effect on abundance in soy landscapes. Duration of substantial human landscape modification had a positive impact on total abundance in maize and rice landscapes.

The variance explained by fixed effects (marginal pseudo-R^2) for all models varied between 0.05 and 0.36, whereas the variance explained by fixed and random effects together (conditional pseudo-R^2) varied between 0.52 and 0.98 (Nakagawa & Schielzeth 2013). These levels of pseudo-R^2 are typical for models based on datasets collected from multiple source, in which most of the variation is explained by the random effects (Williams et al. 2022). The model diagnostic plots suggest that our models do not fully fit model assumptions (Figure S5 – S21). The use of the negative binomial instead of the Poisson distribution for the species richness models did not substantially improve model behaviour (Figure S5-S12). The estimated effects were similar to the Poisson models, with weaker negative effects for the negative binomial land conversion model for species richness (Figure S13, Table S20). The Bayesian models produced very similar results to the original models (Table S21).

Biodiversity impact of closing yield gaps

When closing yield gaps for all four crops, as estimated for the year 2000, 83% of all grid cells would experience a decrease in species richness. Mean species richness change per pixel globally was − 11.8% and the median was − 10.8%. The decrease in species richness is spread across both tropical and non-tropical areas (Fig. 4a). The small areas of species richness increases are likely due to positive rice yield effects in landscapes with a high percentage of natural habitat and wheat yield effects in landscapes with very low percentage of natural habitat (Figure S24). 63% of all grid cells will experience a decrease in total abundance with an average of + 62.6% change driven by high abundance increases for maize and soybean in tropical areas (Figure S25), and a median of -12.7%. Non-tropical areas are almost all characterised by decreases in abundances (Fig. 4b). For RCAR, 34% of grid cells will experience a negative biodiversity impact and almost all of them are in tropical areas (Fig. 4c). The difference in biomes type patterns are most likely driven by soybean and maize yield RCAR effects (Fig. 3I, j). The average projected RCAR change across all grid cells was + 61.6% with a median of 0%. Almost 20% (19.8%) of grid cells had a 0% change in RCAR, these likely representing non-tropical landscapes with wheat cultivation (Figure S26).

Farmland expansion vs intensification

When comparing hypothetical options for increasing total production by 1% in existing agricultural landscapes, neither expansion, nor intensification had uniformly better estimated biodiversity outcomes in all locations. Intensification was associated with better biodiversity outcomes in terms of species richness than farmland expansion on 0.4%, 67.3%, 10.9% and 47.4% of the cultivated areas for maize, soybean, wheat and rice respectively (Fig. 5a, b, c, d). Farmland expansion was associated with higher species richness on the rest of the cultivated areas, i.e., the majority of land cultivated with maize, wheat and rice, especially in non-tropical areas (Fig. 5a, b, c, d). In terms of total abundance, modelled

intensification was associated with more positive outcomes on 37.9%, 34.6%, 5.8% and 55.5% of the cultivated areas for maize, soybean, wheat and rice respectively. Thus, farmland expansion was associated with higher total abundances on most of the land cultivated with maize, soybean and wheat, but intensification was estimated to be better across much of the tropics (Fig. 5e, f, g, h). For RCAR, farmland intensification resulted in better modelled outcomes on 62.3%, 60.5%, 60.4% and 40.2% of the cultivated areas for maize, soybean, wheat and rice respectively. In this case, intensification was the better production-growing strategy compared to farmland expansion for three of the four crops: maize, soybean and wheat. However, farmland expansion resulted in better modelled outcomes in the tropics, especially for maize and soybean (Fig. 5I, j, k, l).

Discussion

When looking at the biodiversity impact of land conversion separately from yield considerations, impacts of both local land use and landscape composition are uniformly negative and relatively strong across species richness, total abundance and relative abundance-weighted community-average range size (RCAR). Our results are consistent with previous studies on local land-use change (Gibson et al. 2011; Murphy & Romanuk 2014; Newbold et al. 2015) and landscape composition effects (Outhwaite et al. 2022). More specifically, our results highlight the importance of considering the landscape context when designing and implementing conservation management. For instance, biodiversity in primary vegetation in agricultural landscapes changes compared to primary vegetation in natural landscapes across all three metrics. This suggests that primary vegetation in human-modified landscapes has less conservation value than primary vegetation in natural landscapes (Chase et al. 2020; Venter et al. 2016) and highlights the high biodiversity costs of expanding agriculture in previously unfarmed landscapes. However, primary vegetation in modified landscapes likely represents an important source of biodiversity and ecosystem services for the surrounding farmland (Karp et al. 2013; Outhwaite et al. 2022).

When looking at the biodiversity impacts of yield or agricultural management, large-scale studies of biodiversity impacts rarely assess how the direction of these effects might differ in different biomes. In our results, tropical landscapes cultivated with maize and soybean show an increase in total abundance and an increase in biotic homogenisation, as measured by RCAR (Fig. 3), suggesting that abundance increases are driven by widespread species (Newbold et al. 2018). Indeed, previous research in tropical areas showed that species with large global ranges are more likely to persist and thrive in agricultural landscapes (Phalan et al. 2011, 2014). In non-tropical soy and maize landscapes, both total abundance and biotic homogenisation generally decrease, suggesting that abundance decreases in these landscapes are also driven by changes in the abundance of widespread species, this time decreases. Supporting evidence for this pattern comes from abundance decreases in common European birds (Inger et al. 2015), often attributed to agricultural intensification (Rigal et al. 2023). There are two considerations that can help interpret these opposing responses of widespread species to yield increases. Firstly, most tropical yields in our data are at the lower end of the yield range whereas non-tropical yields are more equally distributed (Fig. 3, data distribution rugs at the bottom of all panels). Therefore, the tropical trends in our data are likely to be representative of lower yields, whereas the non-

tropical trends likely capture biodiversity trends across a wider yield range. Secondly, non-tropical areas, especially temperate regions, tend to have a longer history of human occupation, and smaller areas of natural habitats than tropical landscapes (Venter et al. 2016). This probably led to farmland-intolerant species being filtered out (Balmford 1996) and farmland-tolerant species increasing in abundance well before the accelerated intensification of the last decades. These ecological communities dominated by farmland-tolerant species in lower intensity agriculture still persist in Eastern European traditional landscapes (Sutcliffe et al. 2015). We speculate that the tropical biodiversity trends in our results capture the type of community change that had happened in the past in non-tropical agricultural landscapes, whereas the non-tropical trends capture mainly the response of farmland tolerant species to the recent high levels of intensification in non-tropical areas.

Previous research has shown the importance of natural habitat in buffering the effects of human activity on biodiversity (Macchi et al. 2020; Outhwaite et al. 2022) but our results suggest nuanced effects depending on crop. In landscapes with wheat cultivation, areas with higher percentage of natural habitat experience steeper decreases in species richness and abundance with yield, likely from higher initial biodiversity as natural habitat is positively related to species richness and abundance in wheat landscapes (Table S20). This suggests that aspects of wheat landscapes or their management might undermine the ability of natural habitat patches to buffer the impact of yield increases, in accordance with research highlighting the importance of landscape context for the biodiversity value of natural habitat patches (Bennett et al. 2004). In maize and soybean landscapes, higher percentage of natural habitat has a positive impact on the effect of yield on total abundance, accelerating increases in tropical and dampening decreases in non-tropical biomes, but a negative biodiversity impact in terms of biotic homogenisation, accelerating RCAR increases in tropical and dampening decreases in RCAR in non-tropical biomes. These results suggest that natural habitat in these landscapes is beneficial mostly to widespread species, that are able to use natural habitat to either benefit or avoid the most serious losses caused by increases in yield (Tscharntke et al. 2002). In landscapes with rice cultivation, higher percentage of natural habitat has a positive impact on the effect of yield on biodiversity as measured by all three metrics. This suggests that species richness and abundance increases with rice yield in landscapes with high percentage of natural habitat is driven by narrow-ranged species. This could be due to the specific requirements of rice cultivation that often involves a period of water submergence (Bambaradeniya & Amerasinghe 2004), which creates temporary wetlands that can be functionally equivalent to more natural wetlands (Elphick 2000). Given that wetland represent only 5–10% of the global land surface and that more than half of the global wetland area has been destroyed or modified (Kingsford et al. 2016), species that are likely to benefit from anthropogenic wetlands created by agriculture are likely to be relatively narrow-ranged (Dudgeon et al. 2006). Our results suggest that a certain amount of natural habitat is necessary for this to happen, as suggested by research that shows the importance of natural habitat for connecting populations of wetland species (Quesnelle et al. 2013).

Although closing yield gaps was proposed over a decade ago as a solution to the challenge of producing food while protecting biodiversity (Foley et al. 2011), there are surprisingly few quantitative assessments of its biodiversity impacts. Compared to existing assessments, our results suggest more severe losses

in local biodiversity when closing yield gaps. Kehoe et al. (2017), who analyse the biodiversity impact of maximising agricultural intensity on all existing farmland, estimate a maximum of 7% loss in species richness, which contrasts with our maximum of 89.1% species loss, and 13% loss is abundance, which contrasts with our maximum 99.7% loss in abundance from closing yield gaps. In their analysis, intensification on all existing cropland leads to little change in species richness and abundance in many areas, compared to our analysis, in which closing yield gaps leads to decreases in species richness on over 80%, and in abundance on over 60% of the global agricultural landscapes. Kehoe et al. (2017) used modelled estimates of biodiversity loss from a global study of land use impact on biodiversity using the same dataset from which we derived our data (Newbold et al. 2015). The difference in results is likely due to the focus of Newbold et al. (2015) on agriculture in general rather than crop type, and absence of interactions between intensity, and natural habitat or biome type in their statistical models (Newbold et al. 2015). As indicated by our analysis, these elements can have opposing effects on biodiversity, and therefore, the negative impacts can be obscured in simpler models. Comparisons with existing assessments of the biodiversity costs of closing yield gaps is complicated further by the use of different biodiversity metrics. For example, Egli et al. (2018) use a metric capturing the value of each pixel with any amount of farmland for the global survival of terrestrial mammals, birds and amphibians. Based on assumed responses of species persistence to agricultural intensification based on habitat preferences, the authors estimate a decrease of 11.1% in the global biodiversity value of agricultural lands due to closing yield gaps. Although the biodiversity metric used by Egli et al. (2018) is not directly comparable to our metrics, the spatial patterns of high biodiversity loss tend to agree, especially with our RCAR patterns in the tropics.

Considering the reduced biodiversity in primary vegetation in modified landscapes (Fig. 2) and the strong association between yield and negative biodiversity outcomes in some landscapes (Fig. 3), it is not surprising that agricultural intensification is not always the best solution for increasing agricultural production in terms of local biodiversity outcomes (Fig. 5). Land-sparing/sharing studies at regional scale usually conclude that land sparing, i.e. intensification, results in better outcomes in terms of reducing extinction risks (Balmford 2021). To the extent that biotic homogenisation of communities as measured here does lead to higher extinction risks, our results disagree with the land-sparing/sharing research as many tropical areas in our results experience higher biotic homogenisation when raising yields by 1% than by expanding cropland by 1% at the expense of primary vegetation. However, our models do not account for the configuration of cropland and primary vegetation areas in our pixels, which is an important aspect of the land-sparing/sharing framework that recommends sharing or sparing of patches of at least 1–10 km^2 (Balmford 2021). Moreover, it is important to emphasise that we are comparing intensification and expansion within existing agricultural landscapes, and that expanding agriculture into new landscapes will lead to significant decreases in biodiversity (Fig. 2).

Our study has several limitations that need to be considered when interpreting the results. First, our focus on the direct association between biodiversity metrics and yields means that relationships may change if agricultural management practices differ from those used at the sampled locations. As a consequence, our results are most representative of the time window represented by the EathStat and

PREDICTS data, which is the time period between 2000 and 2005. Although it is reasonable to assume that agricultural practices have not changed dramatically since then, closing yield gaps would probably result in considerable changes in management practices in some areas, which might change the biodiversity trends presented here. Second, our cropland data were not sampled necessarily within cropland cultivated with the particular crops for which we are using landscape-scale yield estimates. Therefore, our models capture landscape-scale but not local-scale impacts of crop-specific management. Third, our choice of classifying sites into tropical or non-tropical biome types based on latitude, although it avoids other limitations (see *Methods*), leads to unnatural artefacts in our figures where the border between positive and negative impacts is often a straight line (e.g., Figs. 4 and 5), whereas any changes in the relationship between biodiversity and yield in real landscapes would be more gradual. Fourth, all the limitations inherent to the datasets we use, such as geographical and taxonomical biases in PREDICTS (Hudson et al. 2017), and data and downscaling uncertainties in EarthStat (Monfreda et al. 2008) are carried over to our own study. Fifth, there are several limitations of using spatial biodiversity data to infer changes through time, most importantly an inability to consider time lags in biotic changes and responses to environmental changes (Wearn et al. 2012). Finally, some of our more models were less robust to the choice of yield data source (Table S9) and their results should be interpreted with caution.

Both land conversion and increasing agricultural yields have significant biodiversity impacts. In particular, closing yield gaps will likely lead to at a much higher biodiversity cost than previously estimated (Kehoe et al. 2017) and the increase in biotic homogenisation with agricultural yields in the tropics is particularly worrying from a global biodiversity perspective. Given the mostly positive impact of natural habitat on biodiversity and ecosystem services, there is likely a balance that can be struck between intensification and expansion in agricultural landscapes but this balance might change depending on biome, crop, conservation goals and remaining natural vegetation in the landscape. While avoiding expansion of agriculture in unmodified landscapes remains critical for global biodiversity, a reassessment of the approach of closing yield gaps might be necessary, especially in terms of defining a safe intensification levels that can preserve functioning local communities. Hopefully, our results will motivate a renewed focus on important management questions such as how much natural habitat is necessary in agricultural landscapes to maintain biodiversity at sustainable levels (Andren 1994; Arroyo-Rodríguez et al. 2020), how to obtain yield increases with lower biodiversity impacts (Tilman et al. 2011) and how to reduce the overall demand for agricultural products that avoids the hard choice between intensification and expansion (Leclère et al. 2020).

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Figures

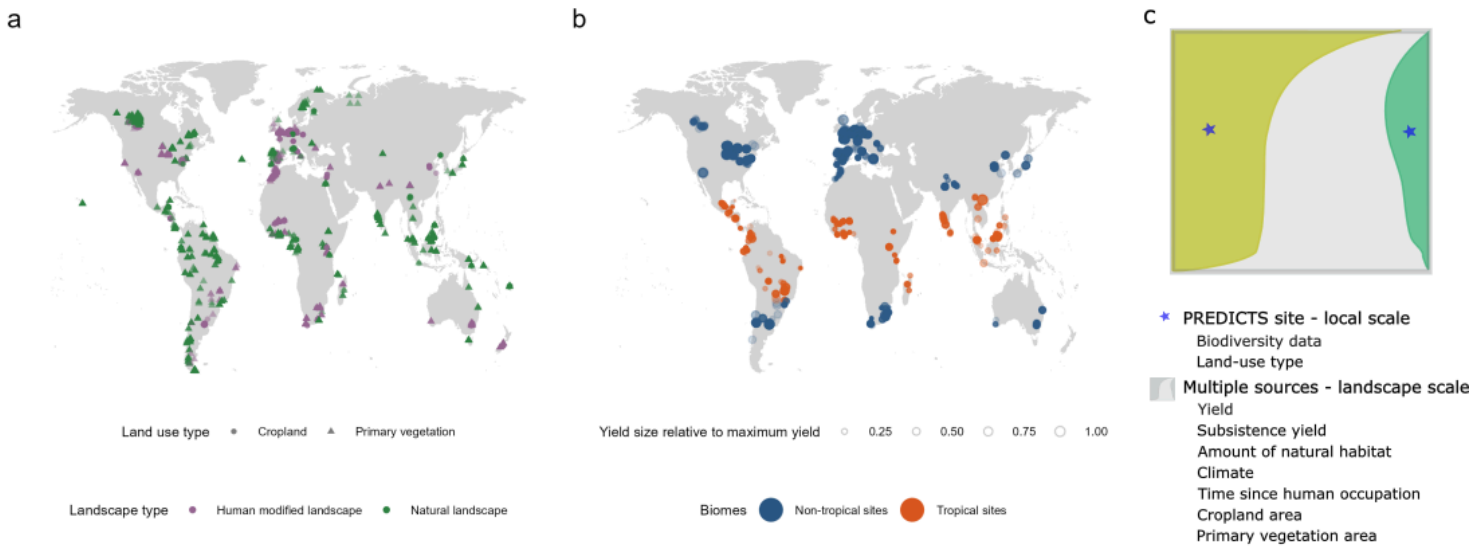

Figure 1

Global distribution of PREDICTS sites used for the land conversion and yield-biodiversity models, and graphical illustration of the spatial relationships between local and landscape-scale variables for the yield-biodiversity models. PREDICTS site distribution for: (a) the land conversion models of biodiversity response to land use and landscape type with circles and triangles showing cropland and primary vegetation sites respectively, and purple and green showing human-modified (less than 30% of natural vegetation) and natural landscapes (more than 70% natural vegetation), respectively; and (b) the yield-biodiversity models of biodiversity response to different yield levels of maize, soy, wheat and rice, with orange and blue indicating tropical and non-tropical sites respectively, and circle size proportional to yield relative to maximum yield for each crop, aggregated across maize, soybean, wheat, and rice. The relative yields were aggregated by weighting each crop yield by the size of the cropland for that particular crop out of the total agricultural area for the four crops in each location. (c) Graphical illustration of the spatial relationships between local and landscape-scale variables for the yield-biodiversity models with the blue stars representing the PREDICTS sites with relevant biodiversity

measurements in cropland (yellow area) and primary vegetation (green area). The grey square represents the scale of the landscape-scale variables.

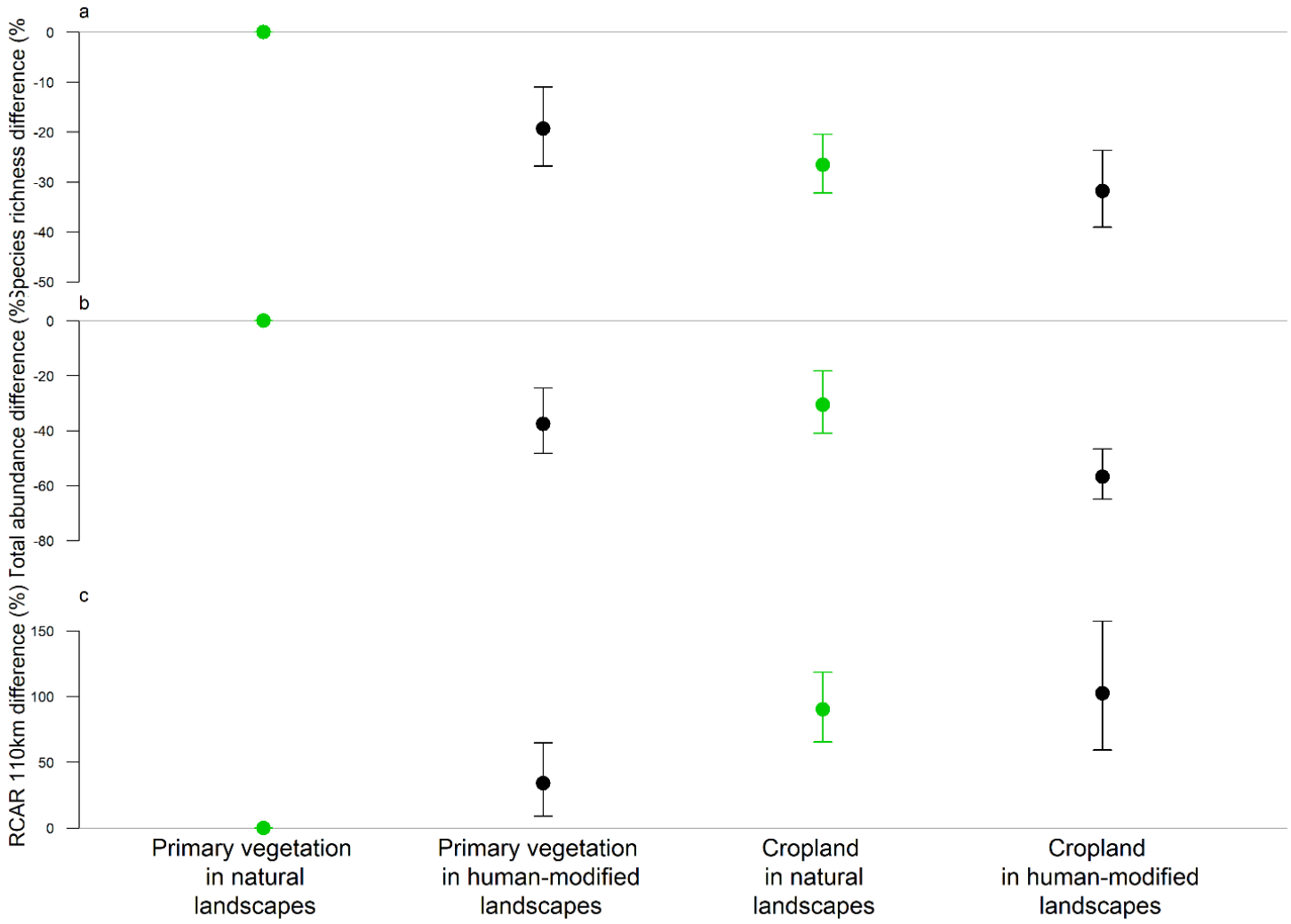

Figure 2

The modelled effect at local and landscape scale of land conversion on three biodiversity metrics: species richness (a), total abundance (b) and relative abundance-weighted community-average range size (c) in primary vegetation and cropland, in natural landscapes (> 70% natural habitat, green dots and error bars) and in human-modified landscapes (< 30% natural habitat, black dots and error bars). Models are based on PREDICTS data sampled in primary vegetation and cropland (10,094 sites from 489 studies). Error bars show 95% confidence intervals.

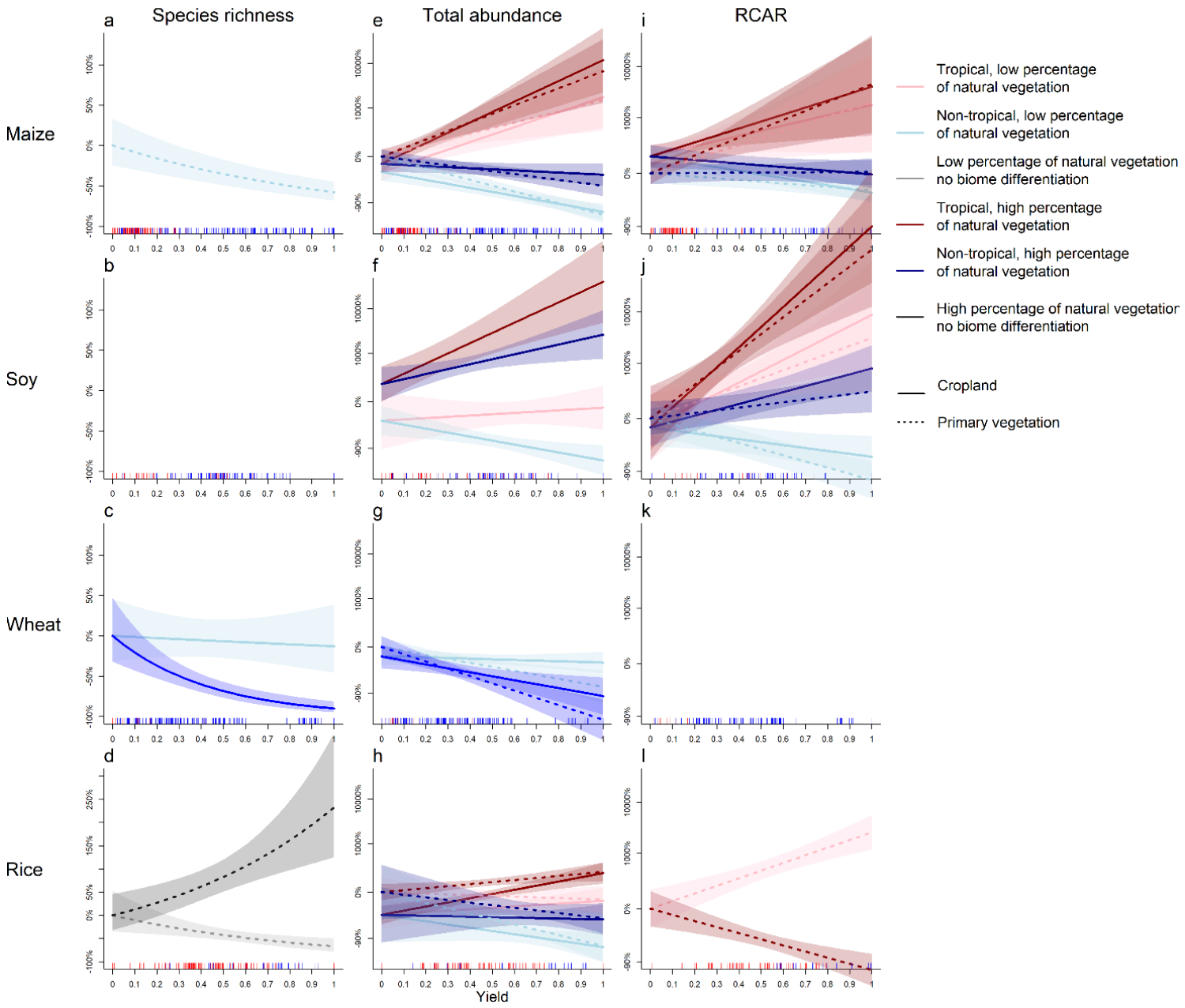

Figure 3

The effect of increasing the yield of four crops on three biodiversity metrics. The plots are organised in rows for each crop: maize (a, e, i); soy (b, f, j); wheat (c, g ,k); rice (d, h, l) and in columns for each biodiversity metric: species richness (a, b, c, d); total abundance (e, f, g, h); relative abundance-weighted community-average range size (l, j, k, l). The modelled biodiversity metrics are based on models using EarthStat estimates of actual yields**.** Species richness and total abundance increases are usually associated with positive biodiversity changes while RCAR increases are associated with negative biodiversity changes (i.e., a homogenisation of community composition). The plots show, where selected in the final model, the responses of biodiversity metrics to interactions between yield and the following variables: biome type–- red hues for tropical biomes, blue hues for non-tropical biomes, and grey hues if biome type was not significant either on its own or in interactions; percentage of natural habitat–- dark

hues for 85% natural habitat and light hues for 15% natural habitat; and land use–- solid line for cropland, dashed line for primary vegetation. Where biome type on its own was selected in the model but the interaction with yield was not selected in the final model, we plotted the values for the biome type with most data points. Where land use on its own was selected in the model but the interaction with yield was not selected in the final model, we plotted the values for the land use type with most data points. Where percentage of natural vegetation on its own was selected in the model but the interaction with yield was not selected in the final model, we plotted the values for low percentage of natural vegetation (15% natural habitat). For cases where yield was not selected in the best model, the plot was left empty. The shaded areas represent 95% confidence intervals. The ticks at the bottom of each plot illustrate the yield values in each dataset rescaled to a [0, 1] interval, with red for tropical biomes and blue for non-tropical biomes. The plots illustrate model predictions that do not necessarily represent plausible combinations of yield and other variable values. The biodiversity metrics were scaled to the level of the respective metric in primary vegetation in agricultural landscapes with the lowest yield among all the sites in the analysis, a value of 0% meaning that the biodiversity level has not changed compared to biodiversity in primary vegetation at the lowest yield. Due to the large range of values, the changes in total abundance (e, f, g, h) and RCAR (I, j, k, l) were $\log_{10}$-transformed to facilitate illustration, and the y-scale is on a logarithmic scale.

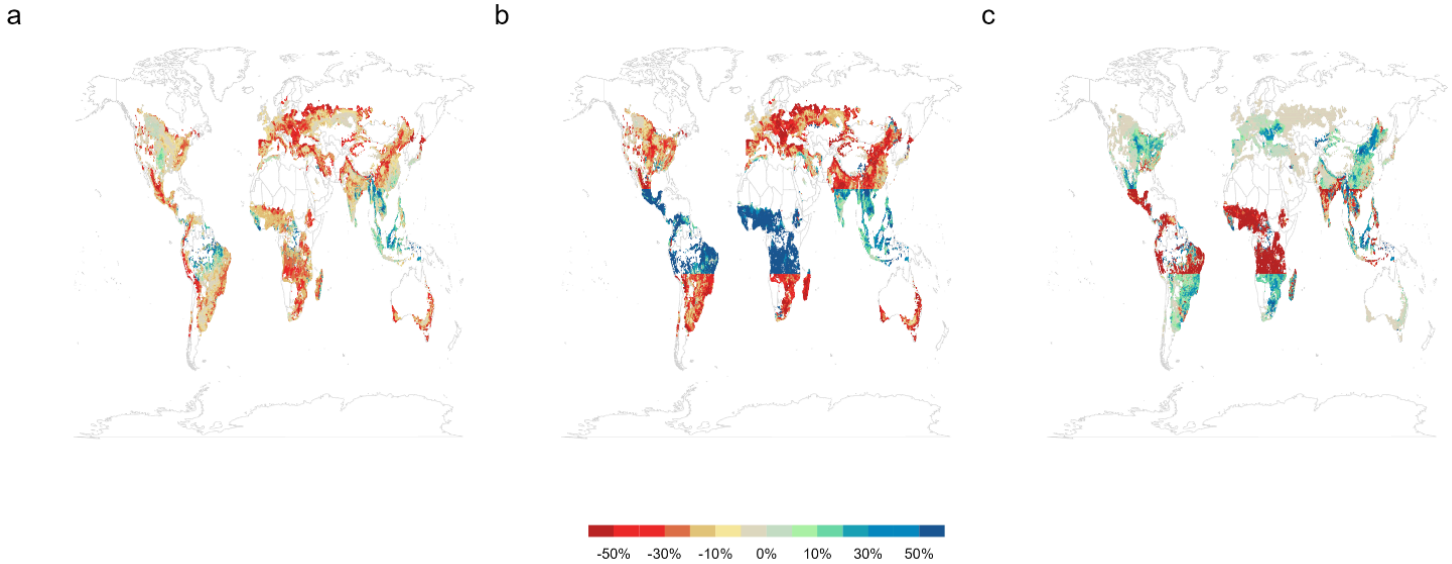

Figure 4

The projected effect of closing yield gaps on three biodiversity metrics: (a) local species richness, (b) total abundance and (c) relative abundance-weighted community-average range size (RCAR), calculated as the percentage difference between biodiversity at yield levels equal to those where yield gaps estimated to have existed in 2000 have been closed, and biodiversity at yield levels estimated for the year 2000. The colour range symbolises positive change (cold hues) and negative change (warm hues) for all plots, including for RCAR percentage changes, which were multiplied by -1. Note that yield benefits from closing yield gaps are spatially variable and depend on the size of yield gaps (Figure S23) and the cultivated area in each pixel (Figure S22).

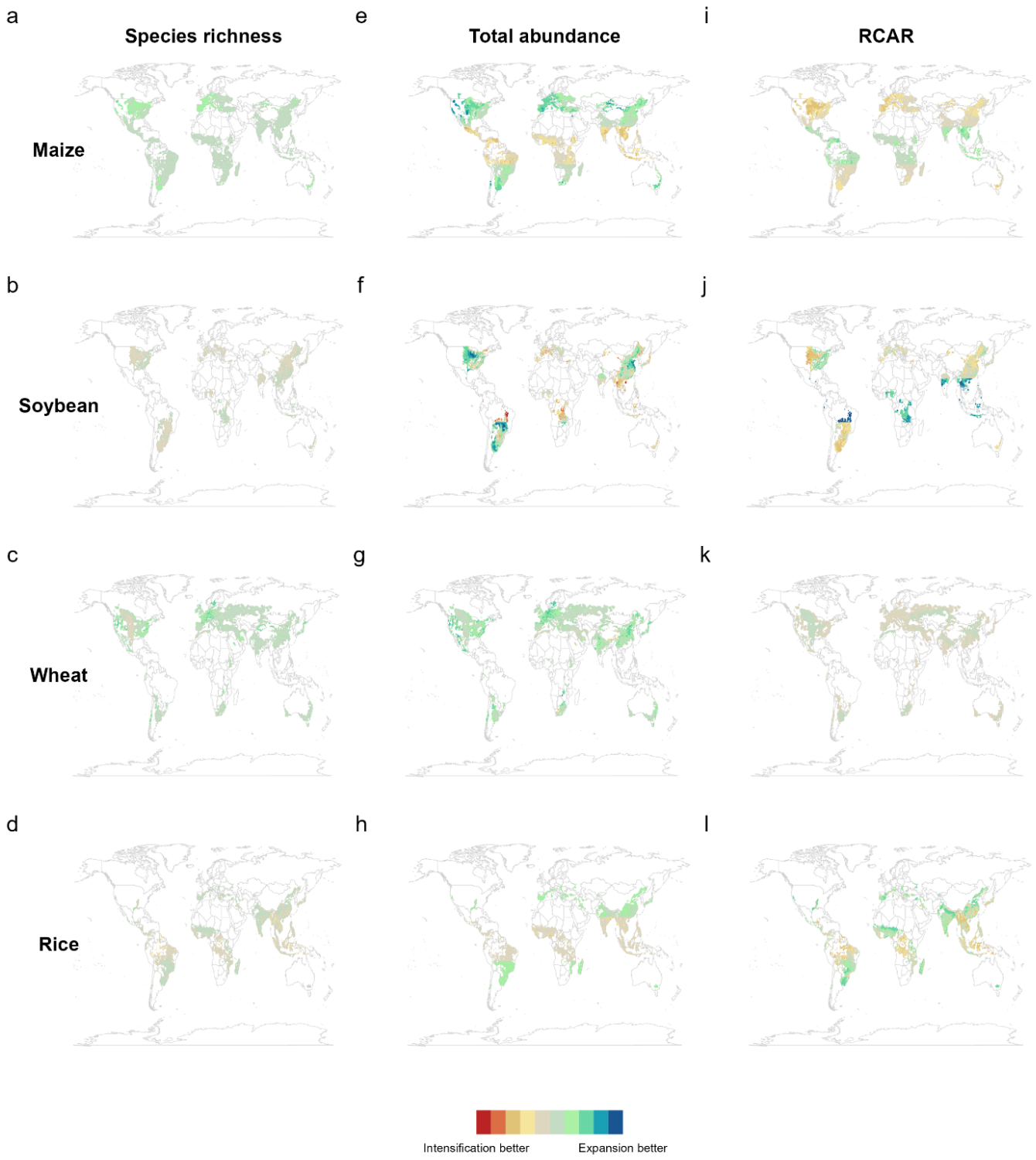

Figure 5

The difference in biodiversity metrics when comparing the land expansion and intensification scenarios. The plots are organised in rows for each crop: maize (a, e, i); soy (b, f, j); wheat (c, g, k); rice (d, h, l) and in columns for each biodiversity metric: species richness (a, b, c, d); total abundance (e, f, g, h); relative abundance-weighted community-average range size (RCAR) (i, j, k, l). The colour range symbolises areas where increasing total production by 1% through cropland expansion is better for biodiversity (cold hues)

and areas where increasing total production by 1% through intensification is better for biodiversity (warm hues). RCAR (i, j, k, l) increases and decreases were considered to be the negative and positive outcomes for biodiversity, respectively.